

Calibrating the Force Sensors

Quick guide — turn raw sensor numbers into real forces (newtons).

1. Why & how (in brief)

Each board reports tiny raw electrical numbers, not forces. The rule that turns them into newtons is

$$\text{force (N)} = (\text{raw} - \text{offset}) \times \text{scale},$$

and its two parts are handled differently. **Calibration** measures the **scale** (a fixed sensor property) by applying **known weights**, and *saves* it. The **offset** (zero point) *drifts*, so it is *not* saved: you set it live with the **Zero Sensors** button (a “tare”) whenever you start measuring.

We use three weights (0, 10, 20 kg): the 0 kg reading is the fit’s zero reference and the loaded readings give the **scale** (slope of the best-fit line) — the third weight makes the fit reliable and confirms the response is linear. Because the wall is **tilted** by an angle θ , one straight-down hang loads both the along-wall (X , by $W \cos \theta$) and out-of-wall (Z , by $W \sin \theta$) axes at once, so a single hang calibrates two axes. The app does all the maths for you.

2. Before you start

- The backend is running, the four sensor boards are **plugged in** (via the USB hub), and the dashboard is **connected** (Connect Device). Captures read live data and time out without a connection.
- You have **known weights** (e.g. 10 and 20 kg) you can hang and pull cleanly on each board.
- You know the wall’s **tilt angle** θ (degrees from vertical; measure it, e.g. with a phone inclinometer).

3. Calibration steps

Open the wizard with **Calibrate** (top toolbar), then:

1. **Start** on the overview screen.
2. **Set the wall angle** θ and continue.
3. **Pick a board** (hands on top, feet on the bottom, as you face the wall). A green check marks axes already done.
4. **Choose what to calibrate:** Hang $\rightarrow X$ & Z (hang weights straight down) or Y — sideways (pull left/right).
5. **Capture each weight.** For each of 0 / 10 / 20 kg: apply the weight, let it stop moving (watch the Steady / Swinging indicator), then click **Capture**.
6. **Review & Save.** The computed scale appears (no offset) — click **Save**.
7. **Repeat** for every board, and for both Hang and Y , then click **Done**.

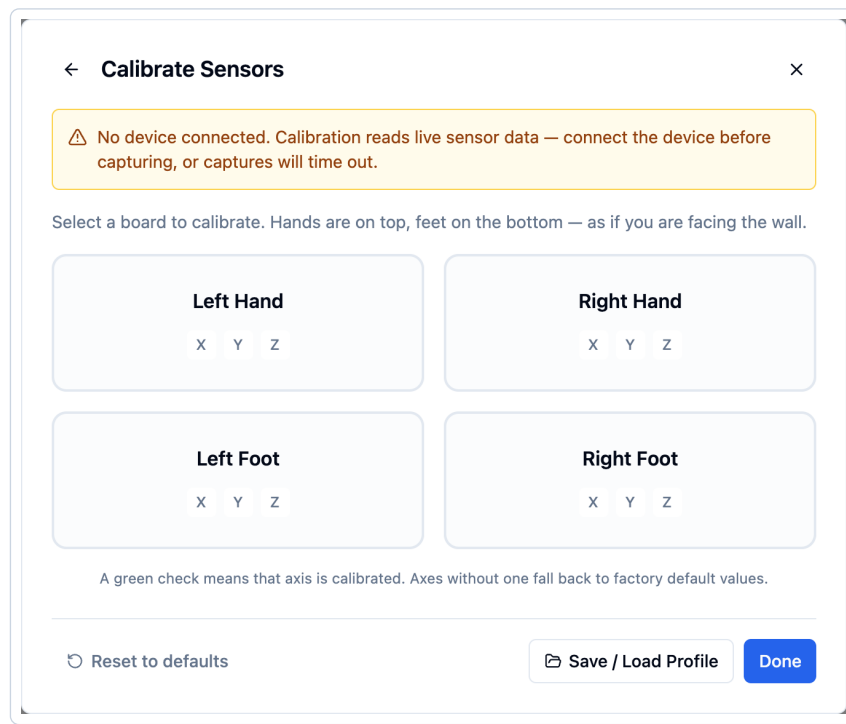


Figure 1. Board picker: calibrate all four boards. Green checks mark calibrated axes; plain letters use factory defaults.

4. Zero the sensors & profiles

Zeroing (each session). With nothing on the holds, click **Zero Sensors** in the toolbar. It averages ~ 1.5 s of live data and re-sets the baseline instantly (recording never pauses). A status chip shows Zeroed X s ago with a green/amber/red freshness dot; re-zero if it drifts. The zero is cached in the browser but *never* saved to the backend, and it needs a live device — demo mode cannot zero.

Profiles. The Profiles button saves the current calibration under a name and reloads it later (e.g. one per wall or sensor set), with **Save as new / Load / Recalibrate / Duplicate / Rename / Delete**. To *change* a profile’s values, use its **Recalibrate** (ruler) button — it loads the profile and opens the wizard, and afterwards **Update “name”** writes the new calibration back. Values are always measured with weights, never typed in. Profiles store scales + axis switches only — never the zero.

5. Good to know

- The angle θ must match the real wall — it only sets how a hang is split into X and Z . A wrong θ gives a wrong calibration. (At $\theta \approx 0^\circ$ a hang cannot calibrate Z , so the wall must be tilted.)
- If a capture is flagged **swung**, the weight was still moving — let it settle and **Re-capture** for a cleaner value.
- Calibration (scales + axis switches) is saved automatically (browser + backend file) and applies immediately. You normally calibrate **once** per board; changing only the display angle (the toolbar’s **View in world frame tilt**, which is separate from the wizard’s θ and never written back to it) does *not* require recalibration. The **zero** is separate and live — set it with **Zero Sensors**, not the wizard.
- Re-zero (and re-click **Connect Device**) after the backend is restarted. Re-plugged sensor boards are picked up automatically.